

Penetration Testing Engineer: Hands-on Experience Course Syllabus

Practical-Oriented Course Syllabus

For CSE Students and Security Professionals

Suggested Duration: 80 Hours | Nature: Theory + Lab + Project

Delivery Scale: Suitable for multiple training batches with approximately 20 trainees per batch

1. Course Rationale

Penetration testing is a critical cybersecurity discipline that simulates real-world attacks to identify vulnerabilities across systems, networks, and applications. This course is designed as a practical, offensive-security-focused training program that develops hands-on penetration testing skills while maintaining strong ethical, legal, and professional boundaries.

2. Target Participants and Delivery Context

This course is intended for:

- Undergraduate and graduate CSE students
- Recent graduates seeking cybersecurity careers
- Security professionals, SOC analysts, and ethical hackers

All practical activities must be conducted strictly in authorized, instructor-controlled laboratory environments.

3. Ethical and Legal Scope

This course emphasizes ethical hacking, responsible disclosure, and strict legal compliance. All testing activities are limited to controlled lab environments, virtual machines, and authorized training platforms.

4. Course Objectives

After successful completion, learners will be able to:

1. Understand penetration testing methodologies and frameworks
2. Perform reconnaissance and information gathering
3. Conduct scanning, enumeration, and vulnerability assessment
4. Execute exploitation techniques in controlled environments

5. Perform post-exploitation and privilege escalation
6. Analyze findings and recommend mitigation strategies
7. Prepare and present professional penetration testing reports

5. Course Outcomes

- CO1. Explain penetration testing lifecycle and methodology
- CO2. Identify vulnerabilities across systems and networks
- CO3. Perform controlled exploitation techniques
- CO4. Conduct privilege escalation and post-exploitation
- CO5. Perform web application attacks based on OWASP standards
- CO6. Recommend mitigation and remediation strategies
- CO7. Deliver professional penetration testing reports

6. Hour Distribution

Theory: 30 Hours

Hands-on Lab: 40 Hours

Project: 10 Hours

Total: 80 Hours

7. Module-Wise Syllabus

7.1 Module 1: Foundations of Penetration Testing and Cybersecurity 6 Hours

Theory focus: penetration testing goals, ethical hacking concepts, threat landscape, attack surface, terminology, legal boundaries, and professional ethics

Hands-on focus: setting up a supervised virtual lab, understanding isolated environments, and preparing a safe penetration testing workflow

7.2 Module 2: Information Gathering and Reconnaissance 8 Hours

Theory focus: OSINT techniques, footprinting, passive reconnaissance, data sources, and attack surface discovery

Hands-on focus: using WHOIS, DNS tools, Shodan, and Maltego for information gathering

7.3 Module 3: Scanning and Enumeration 8 Hours

Theory focus: port scanning, service discovery, enumeration techniques, and protocol analysis

Hands-on focus: using Nmap and enumeration tools to identify live systems and services

7.4 Module 4: Vulnerability Assessment 8 Hours

Theory focus: vulnerability concepts, CVE analysis, risk scoring, and misconfiguration detection

Hands-on focus: running OpenVAS/Nessus scans and validating findings

7.5 Module 5: Exploitation Techniques 10 Hours

Theory focus: exploit fundamentals, payloads, attack vectors, and frameworks

Hands-on focus: using Metasploit and manual exploitation techniques

7.6 Module 6: Post Exploitation and Privilege Escalation 8 Hours

Theory focus: privilege escalation, persistence, lateral movement, and access maintenance

Hands-on focus: credential dumping, privilege escalation, and pivoting techniques

7.7 Module 7: Web Application Security 10 Hours

Theory focus: OWASP Top 10, authentication flaws, input validation, and session management

Hands-on focus: testing SQL injection, XSS, and authentication bypass

7.8 Module 8: Password Attacks and Cracking 6 Hours

Theory focus: password security, hashing mechanisms, brute force and dictionary attacks

Hands-on focus: using Hashcat, Hydra, and John the Ripper

7.9 Module 9: Network Attacks and Sniffing 6 Hours

Theory focus: network protocols, sniffing techniques, and MITM attacks

Hands-on focus: capturing and analyzing traffic using Wireshark

7.10 Module 10: Evasion Techniques 4 Hours

Theory focus: IDS/IPS evasion, antivirus bypass, and obfuscation

Hands-on focus: testing payload evasion techniques

7.11 Module 11: Reporting and Documentation 4 Hours

Theory focus: report structure, risk rating, executive summary, and technical documentation

Hands-on focus: creating professional penetration testing reports

7.12 Module 12: Final Project and Presentation 10 Hours

Theory focus: project planning, scoping, and communication

Hands-on focus: conducting a full penetration test and presenting findings

8. Suggested Laboratory Activities

Includes scanning, exploitation, password cracking, web application testing, packet analysis, and professional reporting.

9. Teaching Methodology

- Interactive lectures
- Instructor-led lab demonstrations
- Hands-on exercises
- Case-based discussions
- Project-based learning

10. Tools and Resources

Kali Linux, Nmap, Metasploit, Burp Suite, Wireshark, Hashcat, OpenVAS

11. Assessment and Marks Distribution

Mid-term Examination: 20%

Final Examination: 25%

Quizzes / Class Tests: 20%

Project / Practical: 25%

Attendance: 10%

12. Suggested Reference Areas

Penetration testing, network security, web application security, incident response, and cloud security

13. Future Development Scope

This course can be expanded into advanced Red Team operations, exploit development, malware analysis, and enterprise-level offensive security training.